UNIT 2 · TOPICS 2.12, 3.2, 4.1 · SAMPLING DISTRIBUTIONS

# Sampling Distributions & Central Limit Theorem

Exact notation, parameter vs. statistic distinctions, standard error formulas, and the conditions required for Normal approximations.

## 1. Variable Definitions: Parameter vs. Statistic

Never mix up **parameters** (fixed, usually unknown numbers describing the entire population) and **statistics** (random variables computed from a sample that vary from sample to sample).

| SYMBOL | NAME | DEFINITION & ROLE | TYPE |
|---|---|---|---|
| $\mu$ | Population Mean | True arithmetic average of all $N$ individuals in the population. | Parameter (fixed) |
| $\bar{x}$ | Sample Mean | Arithmetic average of a sample of size $n$: $\bar{x} = (1/n)\Sigma x_i$. | Statistic (random) |
| $\sigma$, $\sigma^2$ | Population SD / Variance | True spread of individual values in the population around $\mu$. | Parameter (fixed) |
| $s$, $s^2$ | Sample SD / Variance | Sample spread around $\bar{x}$ using $n - 1$ in denominator (Bessel's correction). | Statistic (random) |
| $p$ | Population Proportion | True fraction of successes in the population ($0 \le p \le 1$). | Parameter (fixed) |
| $\hat{p}$ | Sample Proportion | Fraction of successes in a sample: $\hat{p} = X / n$ ($X$ = count of successes). | Statistic (random) |
| $n$, $N$ | Sample / Population Size | $n$ = number of observations in sample; $N$ = total size of population. | Counts ($n \ge 1$) |
| $\mu_{\bar{x}}$, $\sigma_{\bar{x}}$ | Mean & SD of $\bar{x}$ | Center and spread of the sampling distribution of all possible sample means of size $n$. | Sampling Dist. |
| SE | Standard Error | Estimated standard deviation of a statistic computed by plugging in $s$ or $\hat{p}$ when $\sigma$ or $p$ is unknown. | Estimated Spread |

## 2. Sampling Distribution of a Sample Mean ($\bar{x}$)

Suppose we take all possible Simple Random Samples (SRS) of size $n$ from a population with mean $\mu$ and standard deviation $\sigma$.

**CENTER, SPREAD & STANDARDIZATION OF $\bar{x}$**

$$\mu_{\bar{x}} = \mu \quad | \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \quad | \quad \text{SE}_{\bar{x}} = \frac{s}{\sqrt{n}} \quad | \quad z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

**Variables:** $\bar{x}$ = observed sample mean; $\mu$ = population mean; $\sigma$ = population SD; $s$ = sample SD; $n$ = sample size.
**Key Insight:** To cut the standard deviation of $\bar{x}$ in half, you must multiply sample size $n$ by **4**.

**CENTRAL LIMIT THEOREM (CLT)**

Even if the parent population distribution is skewed or non-Normal, as sample size $n$ increases, the sampling distribution of $\bar{x}$ becomes **approximately Normal**: $\bar{x} \sim N(\mu, \sigma / \sqrt{n})$. Standard threshold used in intro/AP stats: $\boldsymbol{n \geq 30}$ (if the population is already Normal, $\bar{x}$ is *exact* Normal for any $n \geq 1$).

## 3. Sampling Distribution of a Sample Proportion ($\hat{p}$)

When counting binary successes ($X \sim \text{Binomial}(n, p)$) and converting to a sample proportion $\hat{p} = X / n$:

**CENTER, SPREAD & STANDARDIZATION OF $\hat{P}$**

$$\mu_{\hat{p}} = p \quad | \quad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \quad | \quad \text{SE}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad | \quad z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$$

**Variables:** $p$ = true population proportion; $\hat{p} = X/n$ = observed sample proportion; $1 - p$ (often denoted $q$) = probability of failure; $n$ = sample size.

## 4. Two-Sample Sampling Distributions (Differences)

For two **independent** random samples of sizes $n_1$ and $n_2$, the mean is the difference of centers, while **variances ALWAYS add** (never subtract variances!).

| STATISTIC | CENTER (MEAN) | TRUE STANDARD DEVIATION ($\Sigma$) | STANDARD ERROR (SE) |
|---|---|---|---|
| **Difference of Means** $\bar{x}_1 - \bar{x}_2$ | $\mu_1 - \mu_2$ | $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ | $\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ |
| **Difference of Proportions** $\hat{p}_1 - \hat{p}_2$ | $p_1 - p_2$ | $\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$ | $\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ |

## 5. Required Conditions Checklist (R-I-N)

| CONDITION | WHAT IT GUARANTEES | HOW TO CHECK (MEANS VS. PROPORTIONS) |
|---|---|---|
| **1. Random** | Unbiasedness: $\mu_{\bar{x}} = \mu$ or $\mu_{\hat{p}} = p$ | Data collected from an SRS, stratified/cluster random sample, or randomized experiment. |
| **2. Independent (10% Rule)** | Validates spread formulas ($\sigma/\sqrt{n}$) when sampling without replacement | Check $n \le 0.10N$ (or $N \ge 10n$). Note: Not needed for randomized experiments with random assignment! |
| **3. Normal / Large Sample** | Normal shape so $z$-table / `normalcdf` applies | **Means:** Population is Normal, OR $n \ge 30$ (CLT), OR sample plot has no strong skew/outliers. **Proportions (Large Counts):** $np \ge 10$ and $n(1-p) \ge 10$ (use $\hat{p}$ or $p_0$ depending on context). |

## 6. Common Traps & Calculator Syntax

**COMMON MISTAKES**

- Using $\sigma$ instead of $\sigma/\sqrt{n}$ when asked about a *sample mean* $\bar{x}$ (read the prompt: one individual vs. average of $n$ individuals!).
- Subtracting variances: $\sigma_{\bar{x}_1-\bar{x}_2} \neq \sqrt{\sigma_1^2/n_1 - \sigma_2^2/n_2}$. Variances **always add**.
- Applying the CLT ($n \ge 30$) to *proportions*. Proportions require the **Large Counts** check ($np \ge 10$, $n(1-p) \ge 10$).

**CALCULATOR & EXAM QUICK CHECK**

- **Probability for $\bar{x}$:** `normalcdf(lower, upper, μ, σ/√n)`
- **Probability for $\hat{p}$:** `normalcdf(lower, upper, p, √(p(1−p)/n))`
- **Unbiasedness:** A statistic is unbiased if its sampling distribution mean equals the parameter ($E(\bar{x}) = \mu$).
- **Precision vs. Accuracy:** Larger $n$ lowers spread (higher precision), while random sampling prevents bias (accuracy).